

Business Intelligence Internship Assignment – Approach & Assumptions

Objective

This project simulates insurance policy sales and claim activity to evaluate the profitability and risk patterns of an auto insurance portfolio. The analysis focuses on premium collection, claims behavior, loss ratios, and potential future liabilities.

Approach

- Generated two datasets: Policy Sales Data (2024) and Claims Data (2025–2026).
- Simulated 1,000,000 policy purchases distributed evenly across all days of 2024.
- Assigned policy tenures based on the defined distribution: 1 year (20%), 2 years (30%), 3 years (40%), and 4 years (10%).
- Calculated premium as \$100 per year of policy tenure.
- Computed policy start date as Purchase Date + 365 days and policy end date as Start Date + tenure.
- Generated claims using business rules: specific purchase dates trigger claims in 2025, while a portion of 4-year policies file additional claims in early 2026.
- Used SQL queries to calculate premium totals, claims breakdowns, ratios, and future liabilities.
- Visualized claim trends using an Excel dashboard with a monthly bar chart.

Assumptions

- All vehicles have the same value: \$100,000.
- Claim amount equals 10% of vehicle value (\$10,000).
- Policies sold uniformly across the year maintain the same tenure distribution daily.
- Each vehicle can file only one claim per year.
- A claim occurs only if the policy is active at the time of the claim.
- Vehicles eligible for claims in 2025 may also file a second claim in 2026.
- Premium revenue is recognized daily across the policy tenure period.

Key Insights from the Analysis

- Total premium collected in 2024 was approximately █240 million.
- Claims in 2025 remained relatively stable each month, averaging around █32–33 million.
- Claims increased sharply in early 2026 due to additional claims from 4█year tenure policies.
- The portfolio loss ratio is approximately 205%, indicating that total claims significantly exceed collected premiums.

Bonus Question Insights

- Most profitable tenure: 3█year policies appear most profitable because they generate higher premium income relative to claims.
- Claim trend dashboard: An Excel dashboard with a bar chart was created to visualize monthly claim trends for 2025 and 2026.
- Portfolio loss ratio: $\text{Claims} \div \text{Premium} \approx 205\%$, suggesting the portfolio currently operates at an underwriting loss.
- If claim frequency increases by 5% annually, total claims could rise to roughly █518 million, pushing the loss ratio above 215%.

Additional Analytical Observations

- Policies with longer tenure generate more stable premium streams but may expose the insurer to extended risk periods.
- Claim concentration on specific purchase dates suggests predictable claim patterns in the simulated dataset.
- Premium recognition analysis shows that only a portion of collected premium is earned by February 2026.
- Improving underwriting policies or adjusting premium pricing could significantly improve profitability.